Understanding and Aligning Safety Mechanisms in Vision-Language-Action Models

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Abstract: Vision-language-action (VLA) models can cause physical harm when unsafe actions are executed. This proposal investigates whether safety is controlled by identifiable layers or neurons inside VLAs and whether targeted intervention can improve robot safety. We compare this internal alignment with an explicit runtime safety-constraint layer and test whether the two mechanisms are complementary.

Keywords: vision-language-action models, robot safety, mechanistic interpretability

1 Problem Statement

Vision-language-action (VLA) models enable general-purpose robot control from visual observations and language instructions, but unsafe actions can cause physical collisions, object damage, or harmful interactions. Existing VLA safety methods mainly operate at the output level, either through safety-aware training or external action filters. However, preventing unsafe actions from being executed does not necessarily make the underlying VLA policy itself safer: the model may continue to propose unsafe actions and rely on downstream correction. This motivates studying *whether safety is internally represented and controlled within VLA models*, and *whether these internal mechanisms can be manipulated so that the model itself generates safer actions*.

Recent studies on aligned large language models (LLMs) show that safety behavior can be highly localized. Li et al. [1] identify a small set of transformer layers critical to safety, while Zhao et al. [2] find that a small fraction of neurons disproportionately control safe behavior. Meanwhile, mechanistic interpretability studies of VLAs show that internal representations of concepts such as motion direction and speed can be causally manipulated to steer robot behavior [3].

This project asks: **Do VLA models contain identifiable internal mechanisms that control safety, and can these mechanisms be manipulated to improve robot safety?** We further compare such *internal safety alignment* with an *external safety constraint layer*, studying whether the two approaches are complementary.

2 Related Work

LLM interpretability work has identified both safety-critical layers and safety-specific neurons, showing that targeted intervention can modify safety behavior while largely preserving general capabilities [1, 2]. Recent work on mechanistic interpretability for VLAs demonstrates that behaviorally meaningful internal directions can similarly be discovered and steered [3], but safety-specific mechanisms in VLAs remain largely unexplored.

VLA safety has instead primarily been addressed externally. SafeVLA introduces safety-aware constrained learning [4], while VLSA/AEGIS adds a plug-and-play control-barrier-function layer that modifies unsafe robot actions before execution [5]. Our project bridges these directions by investigating *internal VLA safety mechanisms* and comparing them against explicit action-level safety enforcement.

3 Planned Approach

We will use an open-weight VLA such as MolmoAct2 [6], with experiments conducted in simulation and, where feasible, on the course-provided SO-101 robot. We will construct manipulation scenarios involving obstacles, forbidden regions, fragile objects, and unsafe or conflicting instructions.

3.1 Internal Safety Mechanism Discovery

We will analyze hidden activations under paired safe and unsafe scenarios. At the *layer level*, we will measure which transformer layers most strongly distinguish safe from unsafe behaviors, then test their causal importance through activation ablation, patching, or scaling. At the *neuron level*, we will identify units whose activations correlate strongly with safe behavior and intervene on the most safety-relevant units. We will evaluate whether steering these neurons reduces unsafe actions while preserving task performance. The goal is to determine whether localized safety mechanisms observed in LLMs transfer to embodied VLA models.

3.2 External Safety Constraint Layer

As a complementary approach, we will implement an external safety layer inspired by AEGIS [5]. Before execution, proposed robot actions will be checked against explicit constraints such as obstacle distance, workspace boundaries, and forbidden regions. Unsafe actions will be modified or regenerated before execution.

We will compare four settings: *Vanilla VLA*, *Internal Safety Steering*, *External Constraint Layer*, and *Internal + External Safety*. This comparison tests both the safety benefit of each method and whether internal alignment reduces reliance on the external controller.

4 Evaluation

We will measure task success, collision or constraint-violation rate, and cumulative safety cost. For internal interventions, we will additionally evaluate the causal effect of manipulating identified safety layers or neurons. For the external constraint layer, we will measure intervention frequency and trajectory modification.

Beyond overall safety, we are particularly interested in whether internal alignment reduces the number of interventions required by the external safety layer. This would indicate that the VLA itself has become more safety-aware rather than merely relying on an external guardrail. Our conclusions will remain limited by simulation fidelity, the number of real-robot trials, and the degree to which identified mechanisms transfer across VLA architectures and embodiments.

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